

30M (0.03B) best pretrain LR by batch size and token budget  
 cell = best LR / its DCLM\_heldout loss; row = chinchilla; wd 0.1

|             | <b>adamw<br/>bs 1M</b> | <b>adamw<br/>bs 2M</b> | <b>adamw<br/>bs 4M</b> | <b>muon<br/>bs 1M</b> | <b>muon<br/>bs 2M</b> | <b>muon<br/>bs 4M</b> |
|-------------|------------------------|------------------------|------------------------|-----------------------|-----------------------|-----------------------|
| <b>0.25</b> | 0.08<br>6.024          | --                     | --                     | 0.014<br>5.478        | --                    | --                    |
| <b>0.5</b>  | 0.04<br>5.392          | --                     | --                     | 0.014<br>5.037        | --                    | --                    |
| <b>1</b>    | 0.04<br>4.909          | --                     | --                     | 0.01<br>4.567         | --                    | --                    |
| <b>2</b>    | 0.014<br>4.555         | 0.028<br>4.906         | 0.08<br>5.230          | 0.02<br>4.444         | 0.014<br>4.539        | 0.01<br>4.893         |
| <b>4</b>    | 0.02<br>4.423          | --                     | --                     | 0.02<br>4.379         | --                    | --                    |
| <b>8</b>    | 0.028<br>4.362         | --                     | --                     | 0.014<br>4.331        | --                    | --                    |
| <b>16</b>   | 0.02<br>4.322          | --                     | --                     | 0.014<br>4.300        | --                    | --                    |
| <b>32</b>   | 0.014<br>4.301         | --                     | --                     | 0.01<br>4.279         | --                    | --                    |
| <b>64</b>   | 0.01<br>4.280          | --                     | --                     | 0.014<br>4.269        | --                    | --                    |